A Machine Learning Approach to Crop-Specific Irrigation Need Prediction in Nepal Using a Random Forest Classifier

Swikrit Subedi
Softwarica College of IT & E-Commerce
Kathmandu, Nepal

***Abstract*—Agriculture is an important part of Nepal's economy, and irrigation plays a major role in crop production. However, in many farming areas, irrigation decisions are still made based on traditional practice or visible crop stress rather than data. This study uses machine learning to predict crop-specific irrigation need as a three-class problem: Low, Medium, and High. A synthetic Nepal-calibrated agricultural dataset from Kaggle was used containing soil, climate, crop, and farm-management features.**

The project followed a structured workflow that included data checking, preprocessing, feature engineering, class imbalance analysis, and model evaluation. Three classification models were tested: Logistic Regression, Decision Tree, and Random Forest. SMOTE was also applied to compare model performance before and after handling the minority class. The models were evaluated using accuracy, macro precision, macro recall, macro F1-score, and confusion matrices.

The results showed that the baseline Random Forest model performed best, achieving a macro F1-score of 0.9678. SMOTE slightly improved the Decision Tree model, but it did not improve Random Forest or Logistic Regression. Overall, the findings show that Random Forest is suitable for this irrigation classification task and can be used as a basis for future irrigation decision-support work in Nepal.

I. Introduction

Agriculture continues to play the central role in Nepal's economy and rural livelihoods. The sector remains closely tied to household income, food security, and regional development, while productivity is strongly influenced by water availability, seasonal rainfall, and field-level management practices [1], [2]. In many parts of Nepal, farmers still decide when to irrigate by following fixed schedules or by checking for signs of crop stress. This can delay irrigation, waste water, and reduce crop productivity [3].

These challenges have become more serious as climate conditions have become less predictable. Irregular monsoon patterns, longer dry periods, and off-season rainfall have made irrigation planning more difficult for farmers in Nepal. This issue is also linked to Nepal's geography, as farming conditions in the Terai plains differ from those in the hill regions. As a result, irrigation demand cannot be managed effectively through a uniform rule-based approach, creating the need for data-driven methods that reflect local soil, climate, and crop conditions [4], [3].

Machine learning is useful for this project because it can learn patterns from several agricultural variables at the same time. In this study, irrigation need was treated as three class classification problem with the labels as Low, Medium and High. The dataset included soil, weather, crop, irrigation, and regional features. Three models were compared: Logistic Regression, Decision Tree, and Random Forest. SMOTE was also tested because the High irrigation need class had far fewer records than the other two classes.

This study presents a reproducible machine learning workflow for irrigation need classification in Nepal. It focuses on main stages of the project including preprocessing, feature engineering, class imbalance handling and final model interpretation using the confusion matrices and feature importance.

II. Literature Review

Machine learning has increasingly been applied to agricultural decision-support tasks such as crop recommendation, irrigation automation, yield prediction, and soil health analysis. Existing studies show that machine learning is useful when agricultural outcomes depend on several factors working together, such as soil, weather, and management practices, rather than simple linear relationships [10], [9]. This is relevant to irrigation because water need is affected by environmental conditions, crop growth stage, and previous management actions.

Recent research further suggests that intelligent irrigation and crop-support systems can improve resource efficiency when environmental variables are combined with data-driven modelling. Previous research shows that machine learning can improve agricultural decision-support systems when it uses meaningful environmental and soil-related data. Studies on adaptive irrigation, crop recommendation, explainable models and engineered soil features have shown the improvements inboth prediction performance and user trust [5], [6], [7], [8].

This study is also based on common methods used in imbalanced classification and tabular machine learning. SMOTE is a widely used method for increasing minority-class samples [11]. However, research on imbalanced data shows that the resampling does not always improves every model so it's effect should be tested rather than assumed [12]. Random Forest is suitable for structured tabular data because it can handle non-linear patterns and also provides feature-importance values [13]. Logistic Regression was included as a simple linear baseline [14]. For this reason, the models were compared

under similar preprocessing conditions, and the effect of SMOTE was evaluated directly.

However, limited work is publicly documented on multi-class irrigation need prediction using a Nepal-calibrated dataset with explicit baseline-versus-SMOTE comparison. This study addresses that gap through a reproducible workflow combining preprocessing, feature engineering, imbalance handling, model comparison, and interpretation.

III. Problem Statement and Objectives

Nepal's agricultural sector depends heavily on seasonal rainfall and local irrigation practices. In many farming areas, irrigation is still done only after farmers notice signs of crop stress. This can result in late irrigation, over-irrigation, or under-irrigation, depending on soil condition, crop stage, and weather. Without a simple data-based prediction method, it is difficult to estimate irrigation demand early, especially across Nepal's different agricultural regions.

This study predicts irrigation need as a three-class classification task: Low, Medium, and High. Soil, climate, crop, and farm-management variables were used for prediction. The main aim was to compare machine learning models and select the most effective one. Since the High class had fewer records than the other classes, SMOTE was also tested to see whether resampling could improve performance.

IV. Dataset Description

The dataset used in this study was a synthetic agricultural dataset from kaggle which was calibrated to reflect Nepal's farming context. The dataset contains a total of 900,000 records which is divided into 630,000 training instances and 270,000 test instances. The training set contained 20 input features and one target variable, *Irrigation_Need*, while the test set contained only the input variables. The target variable was defined as a three-class label with the categories Low, Medium, and High.

The dataset included variables related to soil properties, climatic conditions, crop characteristics, irrigation practice, and regional context. These variables included soil type, soil pH, soil moisture, organic carbon, electrical conductivity, temperature, humidity, rainfall, sunlight hours, wind speed, crop type, crop growth stage, season, irrigation type, water source, field area, mulching use, previous irrigation amount, and region. The dataset was designed to follow Nepal-specific agricultural patterns. It includes traditional seasons such as Barkhe, Hiunde, and Chaite, along with the five historical development regions.

Initial inspection showed that the dataset was clean and well-structured. No missing values or duplicate rows were found in either the training or test set. However, the target variable was imbalanced. In the training set, the Low class represented 58.72% of records, the Medium class represented 37.95%, and the High class represented only 3.33%. This imbalance was important because it increased the difficulty of correctly identifying the minority High irrigation need class during model training and evaluation. Figure 1 illustrates

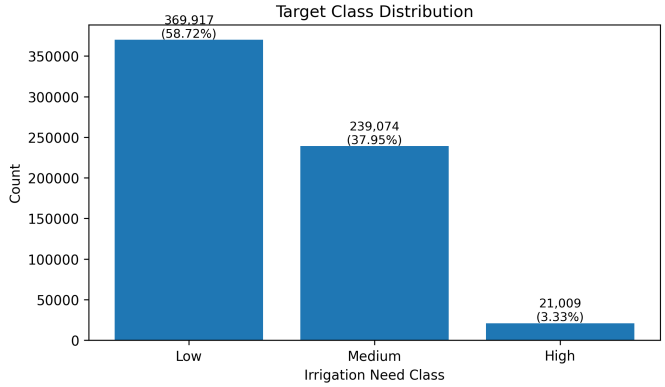

Fig. 1. Distribution of the target classes in the training dataset

the class imbalance in the training data where the Low and Medium classes are more frequent than the High irrigation need class.

V. Methodology

A. Overview

A supervised machine learning workflow was used to classify irrigation need into Low, Medium, and High levels. The main steps were data checking, preprocessing, feature engineering, model training, imbalance handling, evaluation, and final model selection. Three models were compared: Logistic Regression, Decision Tree, and Random Forest. These models were chosen to compare a simple linear model, a single tree-based model, and an ensemble tree-based model.

B. Workflow Summary

The workflow followed this sequence:

$$\text{Raw Data} \rightarrow \text{Validation} \rightarrow \text{Preprocessing} \rightarrow \text{Feature Engineering} \rightarrow \text{Baseline Models} \rightarrow \text{SMOTE Models} \rightarrow \text{Evaluation} \rightarrow \text{Final Model}$$

This process helped keep data preparation, model training, imbalance handling, and evaluation consistent and reproducible.

C. Machine Learning Models

Three classification models were selected for comparison: Logistic Regression, Decision Tree, and Random Forest. Logistic Regression was used as a simple linear baseline. Decision Tree was used because it can identify non-linear patterns and gives results that are easier to understand through rule-based splits. Random Forest was chosen as the final model because it combines several decision trees, gives stable predictions, works well with tabular data, and provides feature-importance values that help with interpretation.

D. Data Preprocessing

The training and test files were first checked for structural consistency, duplicate rows, and missing values. No missing values or duplicate records were found, so imputation and row

removal were not needed. The identifier column was removed because it did not provide useful information for prediction. The dataset was split into training and validation sets using a stratified 80:20 split. This helped keep the class distribution similar in both sets.

Categorical variables were encoded, and numerical variables were standardized inside the preprocessing pipeline. This was important for Logistic Regression and the SMOTE-based experiment, because these methods can be affected by differences in feature scale. For the SMOTE experiment, preprocessing was adjusted to handle both numerical and categorical variables correctly. The same general preprocessing approach was used across the models so the results could be compared fairly.

E. Feature Engineering

Feature engineering was used to create new variables that better captured irrigation-related patterns in the dataset. Four additional features were created:

$$\text{Moisture Deficit} = \text{Humidity} - \text{Soil Moisture} \tag{1}$$

$$\text{Soil Stress Score} = |\text{Soil pH} - 7| + \text{Electrical Conductivity} \tag{2}$$

$$\text{Temperature–Rainfall Ratio} = \frac{\text{Temperature_C}}{\text{Rainfall_mm} + 1} \tag{3}$$

$$\text{Irrigation per Hectare} = \frac{\text{Previous Irrigation_mm}}{\text{Field Area_hectare} + 1} \tag{4}$$

These features were intended to represent water deficit, soil-related irrigation stress, the relationship between atmospheric demand and rainfall input, and irrigation intensity relative to field size. The inclusion of engineered features helped strengthen the practical relevance of the model by embedding domain-informed relationships into the feature space.

F. Model Training and Imbalance Handling

Three classifiers were trained and evaluated. Logistic Regression was used as a linear baseline with a maximum iteration limit of 1000 and class weighting [14]. Decision Tree was limited using depth and split settings to reduce overfitting. Random Forest was trained with 200 trees [13]. In the baseline setting, class weights were used when needed to reduce bias toward the majority class. In the second setting, SMOTE was applied before training to test whether increasing minority-class samples improved performance [11], [12].

G. Evaluation Metrics

Model performance was evaluated using accuracy, macro precision, macro recall, macro F1-score, and confusion matrices. Macro F1-score was treated as the main comparison metric because it gives equal importance to all three classes, including the minority High irrigation need class. The final model was selected based on overall performance and class balance, not accuracy alone.

VI. Implementation and Reproducibility

The project was implemented in Python using both Jupyter Notebook and modular Python scripts. Pandas and NumPy were used for data handling, scikit-learn and imbalanced-learn for model development, and Matplotlib and Seaborn for visualization [15].

The workflow generated model summaries, classification reports, confusion matrices, feature-importance outputs, the final Random Forest model, and test predictions. This structure supports reproducibility and allows the experiments to be rerun from the submitted code and instructions.

VII. Results and Discussion

The results showed that Random Forest performed better than the other models overall. In the baseline setting, Random Forest achieved the best result, with an accuracy of 0.9850 and a macro F1-score of 0.9678. Decision Tree also performed well, with a macro F1-score of 0.9523, while Logistic Regression had a lower macro F1-score of 0.7435. Table I summarizes the performance of all models under both the baseline and SMOTE-based settings. These results indicate that tree-based models were more suitable for this classification task than the linear baseline model.

All precision, recall, and F1 values in Table I are macro-averaged metrics. SMOTE was then tested to see whether handling the minority class improved model performance. The results showed that SMOTE did not help all models equally. Decision Tree improved slightly, with its macro F1-score increasing from 0.9523 to 0.9553. However, Random Forest and Logistic Regression both performed worse after SMOTE. Random Forest decreased from 0.9678 to 0.9637, while Logistic Regression decreased from 0.7435 to 0.6224. This shows that increasing the minority class samples did not always improve the overall model performance.

The class-wise results also showed an important pattern. Logistic Regression had high recall for the High irrigation need class, but its overall balance across all three classes was weaker. Random Forest gave more balanced results across the Low, Medium, and High classes. This explains its higher macro F1-score. Since macro F1-score treats all classes equally, it was used as the main metric for selecting the final model. Based on this, the baseline Random Forest model was selected as the final model.

Overall, the results show that tree-based models worked better for this dataset than the linear Logistic Regression model.

TABLE I
Validation performance of baseline and SMOTE-based models

Setting	Model	Accuracy	Macro Precision	Macro Recall	Macro F1
Baseline	Logistic Regression	0.8430	0.7059	0.8509	0.7435
Baseline	Decision Tree	0.9815	0.9394	0.9671	0.9523
Baseline	Random Forest	0.9850	0.9680	0.9677	0.9678
SMOTE	Logistic Regression	0.7087	0.5893	0.7401	0.6224
SMOTE	Decision Tree	0.9824	0.9476	0.9638	0.9553
SMOTE	Random Forest	0.9833	0.9619	0.9656	0.9637

TABLE II
TOP FIVE FEATURE IMPORTANCES FROM THE FINAL RANDOM FOREST MODEL

Feature	Importance
Soil_Moisture	0.2406
Temperature_C	0.1254
Wind_Speed_kmh	0.0963
Rainfall_mm	0.0706
Crop_Growth_Stage_Harvest	0.0690

Random Forest performed well because it could capture non-linear relationships between soil, crop, climate, and irrigation-related variables. Although SMOTE was useful for comparison, the best result still came from the baseline Random Forest model trained on the original data.

A. Interpretation of the Final Model

Figure 2 shows the class-wise prediction performance of the final baseline Random Forest model. The model performed well across all three irrigation levels. It was especially consistent for the Low and Medium classes, and it also gave strong results for the minority High class. This supports the decision to select Random Forest as the final model.

To better understand the behaviour of the final selected model, feature-importance analysis was carried out using the fitted baseline Random Forest classifier. Table II presents the five most influential predictors identified by the model. The results showed that Soil_Moisture was the most important feature, followed by Temperature_C, Wind_Speed_kmh, Rainfall_mm, and Crop_Growth_Stage_Harvest. These findings are consistent with the agricultural logic of irrigation demand, since water requirement is strongly influenced by existing soil water availability, atmospheric conditions, and crop development stage.

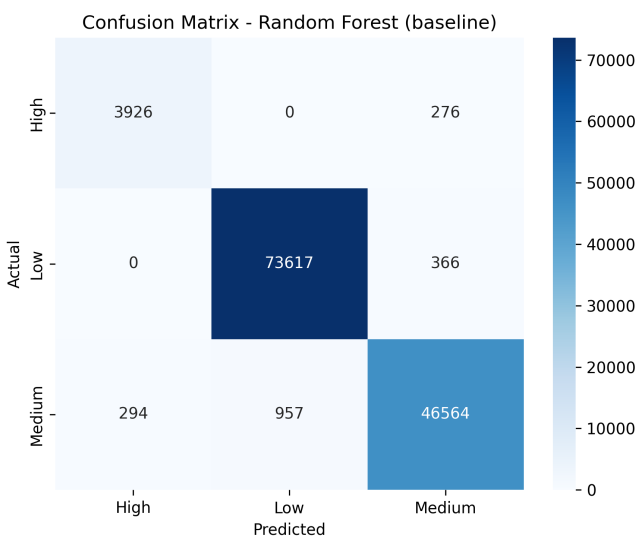

Fig. 2. Confusion matrix for the baseline Random Forest model on the validation set

The strong performance of Random Forest suggests that irrigation need depends on several factors working together, not just one simple relationship. For example, low soil moisture, less rainfall, higher temperature, and dry conditions may have helped the model identify the High irrigation need class more clearly. This also explains why Random Forest performed better than Logistic Regression, since it can handle more complex relationships between features.

B. Practical Implications

The findings show that irrigation prediction should not depend on one factor only. Soil condition, previous irrigation, climate, and crop stage should be considered together. This type of model could help with irrigation planning, support extension workers, and contribute to future decision-support tools for agriculture in Nepal.

VIII. ETHICAL, LEGAL, AND PROFESSIONAL CONSIDERATIONS

Although the final model achieved strong performance, it should be used responsibly. The dataset was synthetic and downloaded from Kaggle, so it may not fully represent real farming conditions in Nepal. Without validation using field data, incorrect predictions could lead to under-irrigation, lower yield, over-irrigation, or water waste.

Professionally, the model should be treated as a decision-support tool rather than a replacement for farmer judgment or agricultural expertise. Any real deployment should consider data quality, transparency, local validation, and fair access for smallholder farmers.

IX. LIMITATIONS

Several limitations should be acknowledged. First, the dataset was synthetic and was downloaded from Kaggle rather than collected directly from field observations. Although the dataset was designed to reflect Nepal's agricultural context, it may not fully capture real-world variability, measurement noise, and local farming conditions.

Second, crop-region patterns and field-size distributions may not fully match actual smallholder farming patterns in Nepal. This may affect generalisability, especially in areas where irrigation behaviour depends on local resource constraints.

Finally, the dataset had no temporal structure, so seasonal irrigation changes could not be modelled directly. Economic factors such as pumping cost, fuel use, and water availability were also not included, although these factors can affect real irrigation decisions.

X. CONCLUSION

This study presented a machine learning approach for predicting irrigation need in Nepal as a three-class classification problem. Three models, Logistic Regression, Decision Tree, and Random Forest, were developed and evaluated using a structured preprocessing pipeline. The models were tested in two settings: baseline training and SMOTE-based imbalance

handling. Random Forest achieved the strongest overall performance, with the highest macro F1-score among all evaluated models.

The comparison also showed that SMOTE did not help every model in the same way. It slightly improved the Decision Tree model, but Random Forest and Logistic Regression performed worse after resampling. Based on the overall results, the baseline Random Forest model was selected as the final model. This study provides a reproducible workflow for irrigation need prediction in Nepal, compares baseline and SMOTE-based models, and interprets the final Random Forest model using a confusion matrix and feature importance. Overall, Random Forest gave the best balance across all three classes for this classification task.

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Appendix

GitHub Repository: nepal_irrigation_ml
Video link: nepal_irrigation_ml_demo